

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Examination Summer 2024

Course: B. Tech.

Semester :VII

**Branch : Electronics Engineering/ Electronics and Communication Engineering/
Electronics and Tele-communication Engineering**

**Subject Code & Name: BTHM705_EX Engineering Economics and Financial
Mathematics**

Max Marks: 60

Date: 10/07/2024

Duration: 3 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	CO	Marks
Q.1 Solve Any Two of the following.		
A) Mention a basic steps for value engineering?	CO1	6
B) Explain different types of Efficiency in Engineering Economics?	CO1	6
C) Discuss different types of inflation?	CO1	6
Q.2 Solve Any Two of the following.		
A) Explain in detail the different element of cost?	CO1	6
B) A Transport company has been looking for a new tyre for its truck and has located the following alternatives:	CO1	6

Brand	Tyre warranty (month)	Price per tyre (R.s.)
A	12	1,200
B	24	1,800
C	36	2,100
D	48	2,700

If the company feels that the warranty period is a good estimate of the tyre life and that a normal interest rate (compound annually) of 12% is appropriate, which tyre should it buy?

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| C) A diesel engine was installed 10 years ago at a cost of Rs. 50,000. It has a present realizable market value of Rs. 15,000. If kept, it can be expected to last five years more, with operating and maintenance cost of Rs. 14,000 per year and to have a salvage value of Rs. 8,000 at the end of the fifth year. This engine can be replaced with an improved version costing Rs. 65,000, which has an expected life of 20 | CO1 | 6 |
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years. This improved version will have an estimated annual operating and maintenance cost of Rs. 9,000 and ultimate salvage value of Rs. 13,000. Using an interest rate of 15%, make an annual equivalent cost analysis to determine whether to keep or replace the old engine.

Q.3 Solve Any Two of the following.

- A) Explain Revenue Dominated Cash flow diagram? CO1 6
- B) The Management of a company finds that while the cost of making a component part is Rs.10, the same is available in the market at Rs.9 with an assurance of continuous supply. Give a suggestion whether to make or buy this part. Give also your views in case the supplier reduces the price from Rs.9 to Rs.8. The cost information is as follows: CO1 6

Particulars	R.S.
Material	3.50
Direct Labour	4.00
Other variable expenses	1.00
Fixed expenses	1.50
Total	10.00

- C) Alpha industry is planning to expand its production operation. It has identified three different Technology for meeting the goal. The initial outlay and annual revenues with respect to each of the technologies are summarized in table. Suggest the best technology which is to be implemented based on the present worth method of comparison assuming 20% interest rate, compound annually. CO1 6

	Initial outlay(Rs.)	Annual revenue (Rs.)	Life(years)
Technology1	12,00,000	4,00,000	10
Technology2	20,00,000	5,00,000	10
Technology3	18,00,000	6,00,000	10

Q.4 Solve Any Two of the following.

- A) Differentiate straight line method of depreciation and declining balance method of depreciation? CO1 6

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| B) Write a short note on Preventive Maintenance? | CO1 | 6 |
| C) Two mutually exclusive projects are being considered for investment. Project A1 requires an initial outlay of Rs. 30, 00,000 with net receipts estimated as Rs. 9, 00,000 per year for the next 5 years. The initial outlay for the project A2 is Rs. 60, 00,000, and net receipts have been estimated at Rs. 15, 00,000 per year for the next seven years. There is no salvage value associated with either of the projects. Using the benefit cost ratio, which project would you select? Assume an interest rate of 10%. | CO1 | 6 |

Q. 5 Solve Any Two of the following.

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| A) Write down the cause of Equipment breakdown and Give disadvantages of Breakdown Maintenance? | CO1 | 6 |
| B) Two years ago, a machine was purchased at a cost of Rs. 2,00,000 to be useful for eight years. Its salvage value at the end of its life is Rs. 25,000. The annual maintenance cost is Rs. 25,000. The market value of the present machine is Rs. 1,20,000.

Now, a new machine to cater to the need of the present machine is available at Rs. 1,50,000 to be useful for six years. Its annual maintenance cost is Rs. 14,000. The salvage value of the new machine is Rs. 20,000. Using an interest rate of 12%, find whether it is worth replacing the present machine with the new machine. | CO1 | 6 |
| C) What is process planning and give steps for the same? | CO1 | 6 |

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